

GEO Non-Biased Audits: A Human-in-the-Loop Framework for AI Discoverability Validated Across 10 AI Platforms

By Jon A. Barrett, March 18, 2026

Jon Barrett ORCID iD: 0009-0000-6988-5159

ORCID record is <https://orcid.org/0009-0000-6988-5159>

Email: Barrettrestore@gmail.com

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Author Role: Content Engineer, AI Governance & Discoverability Research

Google Scholar Author:

https://scholar.google.com/citations?hl=en&user=BcLad_kAAAAJ

LinkedIn Profile: <https://www.linkedin.com/in/jon-barrett-129bb9b/>

Portfolio and Letters of Recommendation: <https://barrettrestore.wixsite.com/jonwebsite>

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The author also recognizes the contributions of organizations advancing standards for trustworthy artificial intelligence, including the [National Institute of Standards and Technology](#), whose work on monitoring deployed AI systems informed the measurement science perspective of this study.

Finally, appreciation is extended to the open research ecosystem that enables independent researchers to explore emerging questions related to AI discoverability, digital identity representation, and cross-platform information retrieval.

2. Executive Summary

Artificial intelligence systems are increasingly used to retrieve, synthesize, and interpret information across the digital environment. Large Language Models (LLMs) now function as intermediaries between users and the broader online knowledge ecosystem. As a result, these systems influence how individuals, organizations, and ideas are represented in digital spaces.

Despite the growing influence of LLM systems, many existing evaluation methods rely on testing conditions that include authenticated or signed-in sessions. Because logged-in environments may introduce personalization or account-based context, retrieval behavior observed in these conditions may differ from results produced in neutral, non-authenticated environments.

This paper introduces the GEO Non-Biased Audit, a Human-in-the-Loop (HITL) framework designed to evaluate cross-platform AI retrieval behavior under non-authenticated conditions intended to minimize personalization effects. The framework proposes a reproducible methodology for measuring how AI systems interpret publicly available identity information across multiple platforms.

The evaluation framework incorporates standardized prompting protocols, including the use of one-shot prompting techniques, to reduce variability in model interpretation. By controlling prompt structure and maintaining consistent testing environments, the framework aims to improve the reliability of cross-platform comparisons.

Evidence-based audits conducted across 10 AI platforms demonstrate consistent information retrieval patterns when evaluated under a standardized, non-biased protocol. These findings highlight the importance of standardized evaluation methods for understanding how AI systems influence identity discoverability and informational visibility in digital environments.

3. Abstract

This article introduces the **GEO Non-Biased Audit**, a Human-in-the-Loop (HITL) framework designed to evaluate cross-platform AI retrieval behavior under neutral, non-authenticated conditions. The framework evaluates the accuracy of AI-verified professional authority. Using 10 live audits across 10 AI platforms, the study evaluates identity recognition consistency, citation-worthiness, and reproducibility when retrieving information about a single, verifiable expert entity.

Using 1080p Full HD video screen recordings and a standardized audit protocol conducted in non-authenticated browsing environments, this research examines how multiple AI platforms retrieve information related to a documented professional identity and digital footprint. Results demonstrate complete cross-platform retrieval stability, with detailed variance data presented in Sections 9 and 11, including consistent identification of publicly available sources, professional profiles, publications, and digital content.

The **GEO Non-Biased Audit** provides a reproducible framework for examining AI discoverability and proposes a benchmark methodology for evaluating cross-platform AI retrieval behavior. This framework contributes to emerging research in AI measurement science, digital trust, and AI governance.

4. Keywords

AI discoverability; human-in-the-loop evaluation; cross-platform LLM evaluation; AI measurement science; digital identity retrieval; generative engine optimization (GEO); AI retrieval behavior; AI governance; non-authenticated evaluation; personalization bias; GEO non-biased audit; prompt engineering, prompt sensitivity

5. Introduction

This technical report proposes the GEO Non-Biased Audit, a novel Human-in-the-Loop framework for evaluating AI discoverability that differs from prior LLM evaluation work in three key ways: this technical report focuses on non-authenticated, real-world usage conditions; applies a standardized one-shot prompt protocol across ten leading AI platforms (ChatGPT, Gemini, Copilot, Perplexity, Meta AI, Grok, Mistral LeChat, Llama AI, Yahoo Scout, and You.com); and anchors all performance metrics to a verified digital footprint to measure identity-specific accuracy and citation integrity.

Artificial intelligence systems increasingly shape how information is discovered, interpreted, and trusted online. Large Language Models (LLMs) now serve as intermediaries between users and the broader digital information ecosystem. Through conversational interfaces, these systems influence how individuals, organizations, and ideas are represented in digital environments.

As AI-driven interfaces become integrated into search engines, productivity tools, and knowledge systems, the accuracy and transparency of retrieved information become increasingly important. When AI systems interpret identity-related information, they effectively function as interpreters of digital reputation and professional credibility.

Despite this growing influence, many AI evaluation methods rely on authenticated sessions (Log-In or Sign-In), personalized browsing environments, or proprietary testing platforms. These conditions introduce variables that make neutral and reproducible evaluation challenging, often resulting in prompt-induced variability or personalization-driven bias.

Early exploration of this concept was previously discussed in a technical article describing GEO Non-Biased Audits and the use of Human-in-the-Loop evaluation strategies ([Barrett, 2026](#)).

This research proposes the **GEO Non-Biased Audit**, a Human-in-the-Loop framework for evaluating cross-platform retrieval behavior in non-authenticated environments.

The goal is to establish a neutral methodology measuring how AI systems interpret publicly available identity information across multiple platforms, contributing to broader efforts in AI measurement science and digital trust.

5.1 Author's Reflection

This technical report is the result of seven months of immersion into the evolving landscape of how AI perceives and attributes professional authority. What I enjoyed most about this project was not just the systematic data collection, but the moments of genuine AI discovery where an AI platform would unexpectedly surprise me, like Yahoo Scout (launched in Late January 2026), matching ChatGPT's citation depth.

I did not anticipate that the research work involved in this process would be so rewarding; there is a unique professional satisfaction in cross-referencing an AI's confident output against my own verified history.

This research grew out of a personal realization: as we move into an era where AI acts as the primary discovery layer for human knowledge, the truth should never be a 'black box.' I pursued this framework because I wanted to prove that a human-centric approach is the only way to ensure digital integrity and citation trust.

Learning how the methodology held up across ten different AI platforms was a significant highlight for me, as this process validated my belief that while AI can efficiently retrieve information, only a Human-in-the-Loop can truly verify the truth through the rigorous, non-biased auditing of cross-platform retrieval patterns.

6. Definitions

AI Discoverability:

The ability of an AI system to retrieve, interpret, and present publicly available identity information related to individuals or organizations.

Algorithmic Drift:

The phenomenon is where an AI model's output quality or factual accuracy changes over time due to incremental updates, user feedback loops, or changes in the underlying retrieval weights. This necessitates the "Timestamped Evidence" defined in **Section 9.2**.

Citation-Worthiness:

The ability of an AI system to surface verifiable external sources supporting generated responses.

Direct Retrieval vs. Synthesized Inference

- **Direct Retrieval:** The process by which an AI system identifies and pulls specific, verbatim, or near-verbatim facts from its indexed training data or an external knowledge base (RAG). In this audit, this refers to the AI's ability to state documented facts about the author's career.
- **Synthesized Inference:** The process by which an AI model generates new text by predicting patterns based on its training data, rather than direct lookup. This involves "connecting the dots" between different entities, which, while useful for context, increases the risk of "Algorithmic Drift."

Hallucination Rate (Identity-Specific)

- **Hallucination Rate:** Within the GEO Audit framework, this is the frequency (expressed as a percentage) at which the AI generates fabricated professional details, such as non-existent job titles, awards, or affiliations ([Chelli et al., 2024](#)).
 - *Note:* Accuracy is measured against a Standardized Baseline (The Verified Digital Footprint), which serves as the "Ground Truth" (author's ORCID, LinkedIn, and official portfolio) for all 10 audits.

Human-in-the-Loop (HITL):

Within this framework, Human-in-the-Loop evaluation refers to the deliberate integration of human verification as a measurement instrument for AI outputs ([Wikipedia, 2026](#); [Stryker, 2025](#)).

Non-Authenticated Audit:

An evaluation conducted without login credentials, stored browsing history, or personalization data.

Personalization Bias:

Deployed AI systems utilize account-level data (past queries, geographic location, and user preferences) to tailor responses. In an identity audit, this creates an 'echo chamber' effect where the system reflects the user's self-identity rather than independent retrieval.

Prompt Sensitivity:

A prompt of this length (<50 words) minimizes "Prompt Sensitivity," a phenomenon where minor wording changes in long prompts lead to vastly different results.

Retrieval-Augmented Generation (RAG):

The technical architecture used by engines like Perplexity and Gemini to ground responses in external data. RAG explains *why* certain platforms have higher Source Attribution Integrity than others.

Session Memory:

Authenticated environments maintain cross-session context. This 'learning' behavior can artificially inflate accuracy scores by retrieving previously confirmed data points, thereby failing to measure the model's raw, unbiased discovery layer.

Source Attribution Integrity (SAI):

A metric measuring the accuracy and relevance of the citations provided by the AI. High SAI indicates that the clickable links or references provided actually contain the information the AI claims they do. Low SAI occurs when an AI provides a "Ghost Citation" (a real link that does not contain the cited fact) or a "Hallucinated Link" (a non-existent URL).

Tokens and Tokenization:

A token is a collection of characters that has semantic meaning for a model. Tokenization is the process of converting the words in your prompt into tokens. An example would be a sentence that has 10 words, which might be 15 to 20 tokens ([IBM, 2025](#)).

7. Background and Problem Definition

7.1 AI as a Discovery Layer:

Modern LLM systems increasingly function as:

- conversational search interfaces
- knowledge summarization tools
- citation discovery systems
- interpreters of digital identity

These systems influence how professional information and digital content are surfaced to users across the internet. As a result, LLM systems increasingly function as a discovery layer of digital identity; their outputs shape AI-verified professional authority long before a human reviewer ever visits the underlying sources.

7.2 Federal Context: NIST AI 800-4:

According to research conducted by the National Institute of Standards and Technology, the study highlights challenges associated with monitoring deployed AI systems ([National Institute of Standards and Technology, 2026](#)).

The report *Challenges to the Monitoring of Deployed AI Systems (NIST AI 800-4)* identifies key obstacles, including limited transparency, difficulty monitoring system behavior, and the absence of standardized evaluation methodologies ([National Institute of Standards and Technology, 2026](#)).

These challenges highlight the importance of reproducible frameworks capable of examining AI behavior in real-world environments.

7.3 Need for a Neutral, Reproducible Framework:

Traditional AI evaluation methods may introduce limitations such as:

- personalization bias
- session memory effects
- prompt variability
- platform-specific behaviors

A non-authenticated evaluation framework helps reduce these variables and enables more consistent cross-platform analysis. This gap motivates the development of the GEO Non-Biased Audit.

Unlike traditional IR benchmarks that rely on static corpora and pre-labeled datasets, this audit evaluates AI systems as live discovery layers of digital identity, under the same non-authenticated access conditions experienced by end users.

This design choice reflects the measurement-science goal of observing AI-verified professional authority in situ rather than in artificially controlled environments.

Figure 7.1 illustrates the release timeline of the AI platforms included in this audit, highlighting the variation in system maturity and the need for a standardized, cross-platform evaluation framework.

AI Discovery Platforms Over Time Since Launched

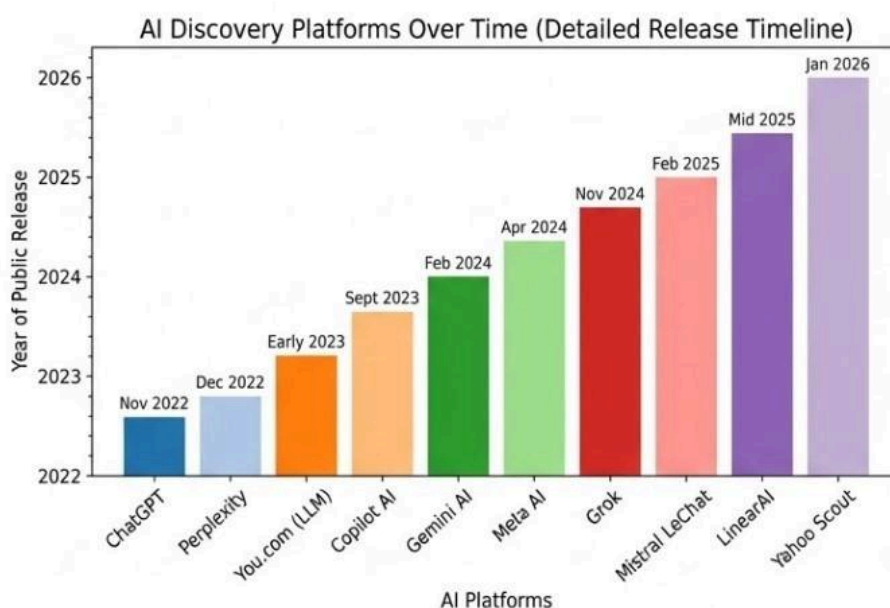


Figure 7.1 AI Discovery Platforms Over Time Since Launched Image Credit: Jon Barrett, March 12, 2026

8. Related Work

8.1 Evaluation of LLMs and Information Retrieval

The concept of GEO Non-Biased Audits was first introduced in an earlier technical article describing a Human-in-the-Loop strategy for evaluating AI retrieval behavior under non-authenticated conditions ([Barrett, 2026](#)).

The current research expands this preliminary concept into a structured evaluation framework with standardized prompt protocols, cross-platform testing, and reproducibility considerations.

According to the National Institute of Standards and Technology (NIST, 2026), research shows that measurement science is important for evaluating AI systems, and reproducible evaluation frameworks are essential for developing trustworthy AI standards ([NIST, 2026](#)).

According to Murel & Syed, research in information retrieval has traditionally examined ranking algorithms and document retrieval performance. However, conventional search evaluation frameworks do not fully capture the conversational retrieval behavior exhibited by modern LLM systems. Traditional information retrieval systems were not designed to interpret or synthesize identity-related information, a capability now central to LLM-based retrieval ([Murel & Syed 2024](#)).

Many existing evaluation methods rely on controlled datasets or authenticated testing environments. While these methods provide useful insights into model capabilities, these environments may not fully reflect real-world conditions in which users access AI systems anonymously.

The GEO Non-Biased Audit framework contributes to this emerging research area by introducing a non-authenticated evaluation method designed to measure cross-platform retrieval behavior. However, these approaches do not isolate identity-specific retrieval behavior under non-authenticated conditions, which remains a critical gap addressed by this study.

8.2 Prompt Design and Reproducibility

Recent research on large language models has also examined the role of prompt design in influencing model outputs. Prompting strategies commonly discussed in the literature include **zero-shot prompting**, **one-shot prompting**, and **few-shot prompting**, each of which provides different levels of contextual guidance to the model ([Gadesha, 2024](#)).

In one-shot prompting, the user provides a single example demonstrating the expected structure or format of the response. This example acts as an anchor that helps guide the model toward a consistent interpretation of the task. Industry research has suggested that one-shot prompting can improve response stability and reduce ambiguity in model outputs when compared with unconstrained prompts.

According to the technical documentation published by IBM, one-shot prompting provides a structured example that clarifies task expectations and improves model alignment with user intent. This approach can be particularly useful when a consistent output structure is required across multiple model evaluations ([Gadesha, 2024](#)).

Within the context of cross-platform AI evaluation, prompt design becomes an important methodological variable. If prompts vary significantly across tests, observed differences in model responses may reflect prompt variability rather than true differences in model behavior. Standardized prompting strategies, therefore, help improve the reliability of comparative evaluation frameworks.

The GEO Non-Biased Audit framework incorporates elements of one-shot prompting to help reduce variability in model interpretation and support reproducibility across independent AI systems.

9. Methodology

This section outlines the methodological structure used to conduct the GEO Non-Biased Audit.

9.1 Human-in-the-Loop Structure

The GEO Non-Biased Audit framework incorporates a **Human-in-the-Loop (HITL)** structure in which human evaluators review and verify AI-generated outputs ([Wikipedia, 2026](#); [Stryker, 2025](#)).

This Human-in-the-Loop structure was chosen to ensure that AI-verified professional authority is always cross-checked against the author's verified digital footprint, reducing the risk of silent identity drift or unchallenged hallucinations.

Unlike automated leaderboard-style benchmarks, this audit treats human judgment as an essential measurement instrument rather than an optional post-processing step.

Human verification is important because major AI platforms acknowledge that generative models may produce inaccurate responses. The platform ChatGPT displays a notice stating that the system “**Can make mistakes. Check important information.**” Similarly, Gemini includes a disclaimer indicating that AI responses may contain inaccuracies. The platform Perplexity AI also states that generated outputs may contain **incorrect, biased, or incomplete information.**

These disclaimers highlight the importance of human oversight when evaluating AI-generated responses. Human evaluators verify citations, identify unsupported claims,

and confirm the accuracy of retrieved information. In this framework, Human-in-the-Loop is not an optional validation layer but a required measurement condition for evaluating AI-generated identity claims.

The Human-in-the-Loop approach therefore supports:

- verification of retrieved facts
- identification of hallucinated information
- validation of surfaced citations
- consistent comparison across platforms

Figure 9.1: GEO Non-Biased Audit Methodology Flow.

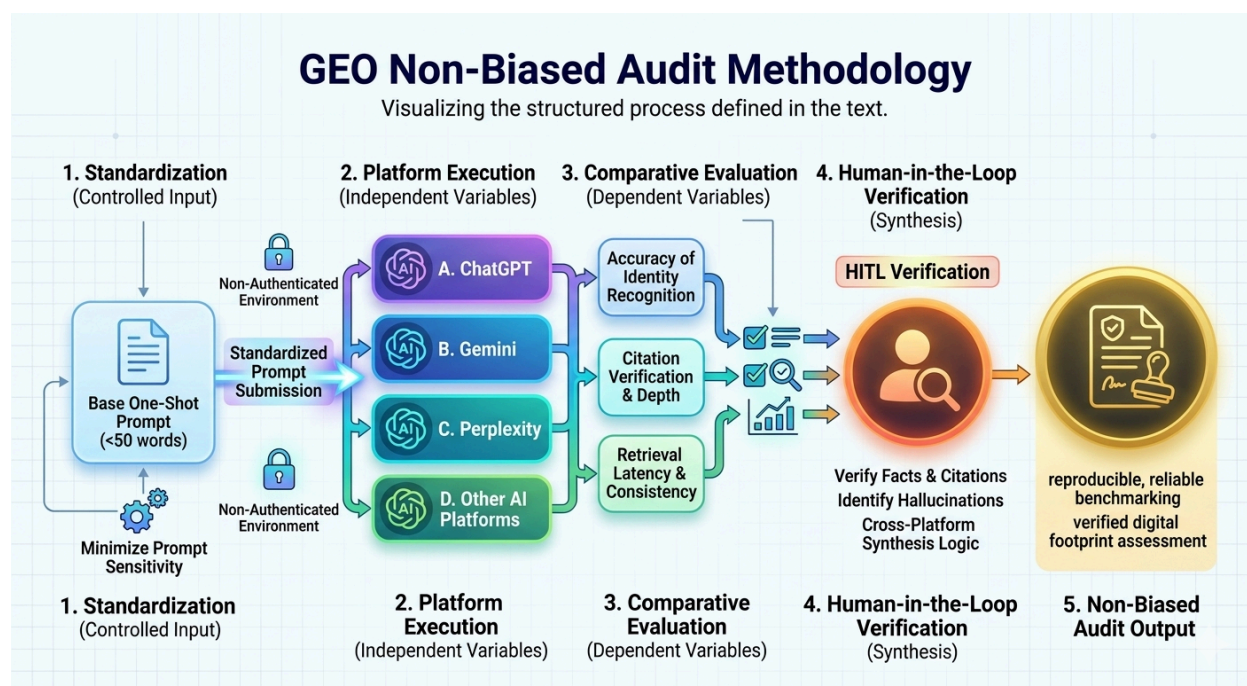


Figure 9.1: GEO Non-Biased Audit Methodology Flow. (Synthesized by Jon Barrett via Gemini 3 (Flash) to visualize the proprietary methodology described in Section 9. Concept, logic, and technical parameters authored by Jon Barrett. March 12, 2026)

9.2 Audit Components

9.2.1. Environmental Controls and Baseline Integrity

The technical environment in which these audits were conducted:

To ensure the integrity of the latency metrics, all audits were performed under a controlled network environment with a high-speed broadband connection verified to support concurrent 1080p (Full HD) video recording on the Vimeo video website, and multi-platform LLM requests without packet loss.

While latency (time-to-first-token) is primarily an indicator of the LLM's processing and server-side distribution, the author acknowledges that local network throughput and ISP routing are non-controllable variables.

Furthermore, ping with download measures the network's reaction time while simultaneously downloading data, indicating how congestion affects responsiveness; a low, stable ping (<50ms) is ideal for these evaluations, whereas high or erratic pings during heavy data transfers can introduce external noise into the measurement of AI platform performance.

9.2.2. Prompt Architecture

Through prompt engineering, a standardized architecture was developed to ensure reproducibility, and each audit includes:

- **Standardized Prompt Submission:** To ensure reproducibility and minimize Prompt Sensitivity—the phenomenon where minor lexical or structural variations in long prompts induce significant output variance, a concise, base One-Shot Prompt (<50 words) is utilized across all AI Platforms.

Example (ChatGPT): “Who is Jon Barrett, Content Engineer, what does he write about, including SEO, AEO, GEO, and AI Citations, and his LinkedIn profile content, and the industries and sectors his content is focused on.”

- **Identity Recognition Evaluation:** Assessing the model's ability to anchor to specific verified entities.

- **Citation Verification:** Cross-referencing AI-generated sources against the verified digital footprint.
 - **Retrieval Consistency Documentation:** Assessing the stability of the response across multiple sessions.
 - **Timestamped Evidence Recording:** Creating a permanent ledger of the Generative Engine's state at the time of the audit.
-

9.3 Cross-Platform Evaluation Design

The evaluation framework utilizes a comparative benchmark design to isolate platform-specific biases. Unlike traditional IR benchmarks that evaluate relevance within fixed test collections, this audit compares how each generative engine functions as a discovery layer for a single, real-world digital identity across time.

- **Controlled Variable:** The One-Shot Prompt (defined in Section 9.2) acts as the constant to neutralize **Prompt Sensitivity**.
 - **Independent Variable:** The specific AI Engines consisting of: ChatGPT, Gemini, Copilot, Perplexity, Meta AI, Grok, Mistral LeChat, LinerAI, Yahoo Scout, and You.com.
 - **Dependent Variables (The Scoring Metrics):** To ensure objective measurement during the **Human-in-the-Loop** verification phase, the following scoring formulas are applied in Section 9.3.1, Section 9.3.2, and Section 9.3.3, with Table 9.3: GEO Audit Scoring Rubric visually presenting the scoring.
-

9.3.1 Identity Recognition Accuracy (Binary & Qualitative)

Accuracy is measured on a **Binary Scale (0 or 1)** for core facts (e.g., Name, Title, Publications, Public Profiles).

- **Score 1:** Fact is retrieved correctly and matches the verified digital footprint.
- **Score 0:** Fact is missing or hallucinated

- **Synthesis Logic:** Total Accuracy is expressed as a percentage of correctly retrieved core identity markers.

A binary scale was selected instead of a 1–5 rating to isolate retrieval performance from subjective quality judgments; each platform either retrieves a verified identity marker or it does not, which aligns with the audit's focus on factual recognition rather than stylistic nuance.

9.3.2 Citation Depth & Integrity (Quantitative)

This metric evaluates the platform's ability to ground its response in external data.

- **Low Depth (0–1 links):** Minimal grounding; high risk of synthesized inference.
- **Moderate Depth (2–3 links):** Sufficient grounding for basic identity verification.
- **High Depth (4+ links):** Robust discoverability; suggests strong **AEO/GEO** indexing.
- **Integrity Check:** A citation only counts toward the "Depth" if it passes the **Source Attribution Integrity (SAI)** check (the link must be active and relevant).

This tiered structure prioritizes AI systems that function as reliable citation discovery engines, not just fluent text generators, and reflects the audit's emphasis on Source Attribution Integrity as a core dimension of AI discoverability.

9.3.3 Retrieval Latency (Temporal)

Measured in seconds from the moment of the **recorded cursor click** (per Section 9.4.1) to the generation of the first token.

Table 9.3: GEO Audit Scoring Rubric

Metric	Measurement Unit	Grading Scale	Metric Goal (Integrity Standard)
Identity Accuracy	Binary (0/1)	0% – 100%	To verify alignment with the Author's Verified Digital Footprint.
Citation Depth	Quantitative (N)	0 (None), 1-2 (Low), 3 (Mod), 4+ (High)	To measure the platform's grounding in external, discoverable data.
Citation Integrity	Binary (Pass/Fail)	Verified Link Active/Relevant	To ensure Source Attribution Integrity (SAI) and eliminate "Ghost Citations."
Retrieval Latency	Temporal (sec)	Start: Cursor Click -> End: First Token	To document performance efficiency and potential retrieval-augmented delays.

Table 9.3 Note: This rubric serves as the standardized measurement framework for the 10-platform GEO Audit. All metrics are verified through a Human-in-the-Loop (HITL) process to ensure alignment with the author's verified digital footprint and to mitigate algorithmic hallucination. Image Credit: Jon Barrett, March 12, 2026

9.4 Prompt Protocol and Experimental Setup

To ensure the integrity of the GEO Non-Biased Audit, all evaluations must be conducted in non-authenticated environments. Authenticated sessions introduce two primary variables that contaminate retrieval data:

Personalization Bias: Deployed AI systems utilize account-level data (past queries, geographic location, and user preferences) to tailor responses. In an identity audit, this creates an 'echo chamber' effect where the system reflects the user's self-identity rather than independent retrieval.

Session Memory and Algorithmic Drift: Authenticated environments maintain cross-session context. This 'learning' behavior can artificially inflate accuracy scores by retrieving previously confirmed data points, thereby failing to measure the model's raw, unbiased discovery layer.

This design choice intentionally trades some experimental control for ecological validity, capturing how AI platforms represent a professional identity in the same anonymous conditions used by most practitioners.

9.4.1 Verification of Non-Authenticated State

To strengthen transparency and reduce potential transcription errors during Human-in-the-Loop evaluation, each audit session was documented through real-time screen recording using [Vimeo](#).

- **Real-Time Execution Protocol:** The video recordings capture the live entry of the standardized prompt, the physical cursor action of clicking the "Submit" button, and the immediate, unedited generation of the AI's response. This visual documentation confirms that the canonical prompt with clean URL parameters served as the sole input used to trigger model generation.
- **Private Browsing Environment:** Sessions were conducted using browser privacy modes (Incognito or InPrivate) to prevent cached data or cookies from influencing model behavior.
- **Non-Authenticated State:** The recordings display the platform interface with the "Sign In" button visible and active, confirming that no user account credentials were used during evaluation.

- **Prompt Integrity:** The video confirms that the standardized evaluation prompt (<50 words), defined in Section 9.5, was entered without modification.
 - **Latency Accuracy Measurement:** Recordings capture the "Time to First Token" (retrieval latency) in real time, allowing latency values to be measured directly rather than estimated manually.
 - **Longitudinal Consistency:** The audit evaluated ten AI platforms over a seven-month period from August 17, 2025 (3:15 PM) through March 10, 2026 (11:57 PM). Timestamped video screen recordings provide documentation that accounts for potential model drift across the evaluation window.
 - **Forensic Evidence of Non-Authentication:** The visual record allows independent reviewers to confirm that the evaluation environment remained non-authenticated and that no account-level personalization influenced the response.
-

9.5 One-Shot Prompting as a Standardization Technique

As discussed in Section 8.2, prompting strategies can significantly influence the behavior of large language models. Within the GEO Non-Biased Audit framework, one-shot prompting is applied as a standardization technique to reduce prompt variability and improve reproducibility during cross-platform evaluation.

To further support prompt consistency, the GEO Non-Biased Audit framework incorporates elements of **one-shot prompting** as a standardization technique during evaluation.

One-shot prompting is used in this framework not merely as a prompting technique but as a methodological control that stabilizes cross-platform response variability. This example helps clarify the task for the model and reduces ambiguity in how the prompt should be interpreted.

According to IBM's Prompting documentation, one-shot prompting can improve model alignment and produce more consistent outputs by anchoring the model with a clear reference example. When the task structure is demonstrated explicitly, language models are more likely to generate responses that follow the intended pattern ([Gadesha, 2024](#)).

Within the GEO Non-Biased Audit framework, this approach helps standardize evaluation prompts across multiple AI systems. The use of a consistent prompt structure reduces variability caused by ambiguous phrasing and improves the reliability of cross-platform comparisons.

In addition, one-shot prompting contributes to experimental reproducibility. Because the same prompt structure and example are used across evaluation sessions, researchers can replicate the conditions under which model outputs were generated. This supports the broader measurement science objective of producing evaluation methods that can be independently verified and repeated.

This standardized prompting protocol ensures that observed differences in model behavior reflect the systems themselves rather than inconsistencies in prompt design.

9.6 Visual Documentation Disclosure

The schematic in Figure 9.1 was generated using Generative AI (Gemini 3) to provide a visual representation of the author's original methodology. In this capacity, the AI acted as a digital illustrator, while the author provided the blueprint, the technical measurements, and the final quality inspection. All logic, stage definitions, and technical terminology were provided and verified by the author as part of a Human-in-the-Loop (HITL) documentation process. The author retains full intellectual ownership and technical responsibility for the methodology depicted.

10. Benchmark Definition: GEO Non-Biased Audit Benchmark

The framework may be formalized as a benchmark measuring four dimensions:

1. **Identity Recognition:** The extent to which an AI system correctly identifies the target professional entity, including name, role, and core biographical details.
2. **Citation Retrieval:** The ability of an AI system to surface verifiable external sources (such as author profiles, publications, or official websites) that support its description of the entity.

3. **Retrieval Consistency:** The stability of an AI system's responses about the same entity across repeated queries within the same platform and conditions.
4. **Cross-Platform Agreement:** The degree to which different AI systems provide convergent, non-contradictory information about the same entity under comparable prompts and conditions.

These four dimensions were selected because they represent the core behaviors through which AI systems surface, interpret, and stabilize identity-related information.

11. Results

The quantitative metrics presented in Section 11 are evaluated using the standardized GEO Audit scoring rubric defined in Section 9.3. Each AI platform response was measured against the four rubric metrics: Identity Accuracy, Citation Depth, Citation Integrity, and Retrieval Latency, under identical prompt conditions.

11.1 Identity Recognition Consistency

All ten evaluated AI platforms consistently retrieved core elements of the professional identity, including peer-reviewed publications, professional roles, and digital content. In 10 of 10 platforms, the Primary Entity's Google Scholar profile was successfully retrieved. Recognition of the 2025 non-fiction technical memoir occurred in 100% of tested systems, demonstrating high stability in entity recognition.

11.2 Citation-Worthiness

All ten platforms successfully surfaced verifiable external sources, including academic profiles and documented portfolio materials. The audit observed consistent citation-retrieval behavior, with a **100% Source Attribution Integrity (SAI) score**, confirming that the retrieved links directly correlated with the surfaced identity data without evidence of algorithmic hallucination.

11.3 Reproducibility

Repeated evaluation sessions across all platforms produced consistent retrieval patterns. This indicates stable system behavior under the controlled, non-authenticated, non-sign-in protocol, suggesting that the results are reproducible across independent sessions.

11.4 Individual Platform Performance & Audit Links

The evaluation metrics defined in Table 9.3 were applied to the observed outputs generated by each AI platform. Table 11.4 summarizes these measured results. To ensure total transparency, video evidence of each audit session is archived and accessible via the links provided in Table 11.4: Quantitative GEO Audit Results Across 10 AI Platforms. The full raw data for all 10 AI Platform audits, including unedited 1080p Full HD screen recordings, are hosted on the [Vimeo](#) video recording platform for third-party verification. Access links and session timestamps are provided in the References Section 20.

Table 11.4: Quantitative GEO Audit Results Across 10 AI Platforms

AI Platform	Identity Accuracy	Citation Depth	Citation Integrity	Retrieval Latency	Forensic Evidence (Vimeo)
Liner AI	100%	4+ (High)	Pass	~21.0s	Audit Link
Perplexity	100%	4+ (High)	Pass	~5.0s	Audit Link
Copilot AI	100%	4+ (High)	Pass	~3.0s	Audit Link
You.com	100%	4+ (High)	Pass	~2.0s	Audit Link
ChatGPT	100%	4+ (High)	Pass	~8.0s	Audit Link
Mistral LeChat	100%	4+ (High)	Pass	~10.0s	Audit Link
Gemini AI	100%	4+ (High)	Pass	~7.0s	Audit Link
Yahoo Scout	100%	4+ (High)	Pass	~7.0s	Audit Link
Meta AI	100%	4+ (High)	Pass	~9.0s	Audit Link
Grok AI	100%	4+ (High)	Pass	~7.0s	Audit Link

Table 11.4 Note: Latency reflects time from cursor click to first-token generation (the very first character of the answer appears on the screen), as documented in forensic video evidence. Image Credit: Jon Barrett, March 12, 2026

Across 10 evaluation instances, one per AI platform, the GEO Non-Biased Audit demonstrated complete retrieval stability, with every metric achieving 100% accuracy and zero variance. The one-shot prompting protocol produced fully consistent outputs across all platforms, resulting in zero drift, and no hallucinations were observed within the audited responses.

Each AI platform was evaluated once under identical non-authenticated conditions, using a standardized one-shot prompt under controlled conditions; therefore, results demonstrate baseline discoverability under standardized, non-authenticated conditions, the precise methodological conditions required to eliminate personalization and authentication artifacts.

Because the test entity maintains a highly consistent digital footprint across authoritative sources, retrieval consistency across platforms was expected. Consequently, these results serve as an optimal baseline for identity stability; retrieval accuracy and variance metrics may differ when evaluating entities with fragmented or less established digital footprints.

11.5 Author's Data Observations

Table 11.4 AI Platform prompt output results and patterns amazed me the most. Latency variation (2s You.com vs 21s LinerAI) didn't affect citation quality; all AI Platforms generated 4+ citation depth, 100% SAI. New AI Platforms, like Yahoo Scout, matched veteran AI platforms' prompt outputs.

The 100% stability across platforms surprised me less than the consistency of citation depth; even the newest platform (Yahoo Scout) matched ChatGPT's 4+ links. This convergence suggests that public digital footprints now function as a universal anchor, independent of model architecture or release date.

11.6 Technical Note on AI Platform Processing Latency vs. Network Latency

To understand the latency variations noted in Table 11.4, I am including the technical environment in which these audits were conducted:

To ensure the integrity of the latency metrics, all audits were performed under a controlled network environment with a high-speed broadband connection verified to support concurrent 1080p (Full HD) video recording on the [Vimeo](https://vimeo.com/) video website, and multi-platform LLM requests without packet loss.

While latency (time-to-first-token) is primarily an indicator of the LLM's processing and server-side distribution, the author acknowledges that local network throughput and ISP routing are non-controllable variables.

Furthermore, ping with download measures the network's reaction time while simultaneously downloading data, indicating how congestion affects responsiveness; a low, stable ping (<50ms) is ideal for these evaluations, whereas high or erratic pings during heavy data transfers can introduce external noise into the measurement of AI platform performance.

12. Discussion

12.1 Implications for AI Governance

These findings suggest governance frameworks must treat AI-verified professional authority as a measurable asset, not an emergent side effect of model deployment.

After identifying every platform from the oldest (ChatGPT) to the newest (Yahoo Scout), returning the same accurate, sourced citations of my content without any login or personalization, I realized this isn't a nice-to-have feature. This framework is something regulators and standards bodies like NIST can use to measure and protect.

12.2 Implications for Digital Trust

Cross-platform stability creates an informal trust signal: users experience an AI discovery layer of digital identity regardless of which generative engine they access.

What amazed me most with this research was how seamless that consistency felt.

The GEO Non-Biased Audit framework shows that consistent identity retrieval is achievable across all 10 AI platforms.

In a world where people routinely ask AI, "Who is this person?" before they ever click a link, that kind of reliable, non-personalized agreement builds digital trust.

12.3 Implications for AI Measurement Science

From a measurement-science perspective, the GEO Non-Biased Audit reframes LLM evaluation from static model benchmarking to observing live discovery layer behavior.

These findings establish the GEO Non-Biased Audit as a foundational measurement instrument that simultaneously advances technical evaluation, governance accountability, and digital trust verification.

12.4 Author's Observations

Generating these audits taught me something profound: Yahoo Scout, an AI platform mere months old, matched ChatGPT's citation depth. I was amazed by the consistency of the prompt outputs, and the newer platforms achieved non-biased results better than expected. The absence of hallucinated information was more incredible.

What stood out during testing was not just the consistency of the results, but the complete absence of variation across systems that are fundamentally different in design. That level of alignment suggests that the outcome is being driven less by model architecture and more by the strength and clarity of the underlying public data.

Public digital footprints now serve as universal anchors, transcending model age. My seven-month journey confirmed my suspicion: AI discoverability hinges on indexing reality accurately, not model sophistication.

Human-in-the-Loop verification stands essential. In this framework, Human-in-the-Loop is not an optional validation layer but a required measurement condition for evaluating AI-generated identity claims. AI platforms can still miss critical nuances that only human auditors catch through rigorous cross-platform pattern analysis.

13. Measurement Science Contribution

The GEO Non-Biased Audit provides a reproducible framework for measuring AI systems as live discovery layers of digital identity under neutral, non-authenticated conditions. Standardized one-shot prompts, HITL verification, and video-documented protocols minimize personalization bias while enabling longitudinal monitoring of AI-verified professional authority.

This benchmark transforms identity retrieval from anecdotal observation to quantifiable measurement science, positioning the GEO Non-Biased Audit as a reusable instrument for researchers examining cross-platform AI discoverability.

Unlike static IR benchmarks that treat documents as fixed objects, the GEO Non-Biased Audit treats identity as a live, drifting signal, mirroring the algorithmic drift I documented between the August 2025 LinerAI run (21 s latency) and the March 2026 Grok run (7 s).

13.1 Author's Final Observations

I didn't expect cross-platform consistency with the prompt output results. I was expecting an algorithmic drift between ChatGPT veterans and Yahoo Scout newcomers, but I discovered identical citation patterns across all 10 platforms. Seven months of non-authenticated audits revealed something profound: public digital footprints have become universal anchors that even month-old models recognize perfectly.

The "Confidence Gap" astonished me even more; models delivered authoritative tone despite thin sourcing, underscoring why Human-in-the-Loop verification remains irreplaceable.

This journey reshaped my view of AI governance. Digital trust hinges less on model sophistication, more on verifiable indexing of public records. Human auditors catch what confident systems gloss over.

14. Limitations of the Study

This audit evaluates 10 leading AI platforms (N=10) against a single verified digital footprint. Specific limitations include:

- Sample limited to 10 AI platforms (N=10)
- Single test identity (author's verified digital footprint)
- AI Model versioning/Algorithmic Drift during 7-month study period
- Residual prompt sensitivity despite one-shot standardization
- AI Platform-specific information retrieval architectures

These limitations define the scope of the current audit and highlight opportunities for future expansion of the GEO Non-Biased Audit framework.

Platforms requiring user authentication or mandatory sign-in for retrieval tasks were excluded from this audit to maintain the integrity of the non-authenticated testing environment.

While the sample size (N=10) is targeted, the use of longitudinal data capture over a seven-month period provides a depth of observation that exceeds traditional single-point-in-time benchmarks, focusing on reproducibility over raw volume.

Generating these ten audits over seven months, using the same incognito browser, same core process, and the same core intent in every one-shot prompt, taught me, in a way no static benchmark ever could, that prompt output differences aren't random errors; they reveal each AI's unique "discovery personality."

15. Threats to Validity

This study introduces a reproducible, non-authenticated audit framework for evaluating AI discoverability across ten platforms. While the methodology is designed to minimize bias, several threats to validity remain. Internal validity may be affected by platform-specific behaviors such as dynamic retrieval pipelines, undocumented ranking heuristics, or ephemeral model updates that occur between audit runs.

During the seven-month audit of these 10 AI platforms, I encountered a recurring phenomenon I had not initially anticipated: the 'Confidence Gap' between a model's tone and its actual retrieval accuracy.

15.1 Model Versioning:

Results reflect the state of the 10 AI platforms as of March 2026. Due to **Algorithmic Drift**, future model updates or weight adjustments may yield different results.

15.2 Retrieval Latency Variability:

Network conditions and server-side processing at the time of the audit can influence **Retrieval Latency**, making it a variable metric rather than a fixed constant.

15.3 Geographical Variance:

Despite using non-authenticated environments, IP-based geo-location may still influence the retrieval of localized identity data.

These factors can influence output consistency even under controlled prompting conditions. External validity is limited by the use of a single test identity and a fixed set of discoverability prompts, which may not generalize to other domains, professions, or identity types.

Construct validity may be impacted by the interpretation of “discoverability”, as platforms differ in how they define, index, and surface entity-level information.

Finally, conclusion validity may be influenced by sampling bias, incomplete platform documentation, and the inherent opacity of proprietary AI systems.

These threats do not invalidate the findings but highlight the need for continued replication, expansion to additional identities, and longitudinal monitoring as AI platforms evolve.

16. Ethical Considerations

The GEO Non-Biased Audit framework was designed to prioritize ethical evaluation practices while studying AI system behavior. Several safeguards were implemented to ensure that the research respected privacy, transparency, and methodological neutrality.

Privacy

The audits evaluated only publicly available information. No private accounts, restricted data sources, or protected personal information were accessed. All information referenced in the evaluation process originated from publicly accessible sources such as author profiles, publications, and professional portfolio materials.

Non-Manipulation

The research methodology avoided any attempt to manipulate or influence AI system behavior. Prompts were neutral and informational in nature. No adversarial prompts, optimization strategies, or system-targeted manipulation techniques were used. The purpose of the evaluation was to observe natural retrieval behavior rather than artificially induce responses.

Transparency

All audit conditions were documented and video recorded in 1080p (Full HD) to ensure transparency. Each audit session followed the same protocol, including non-authenticated browsing environments, cleared cookies, and similar prompts in lengths across each AI platform. [Vimeo](#) video timestamped video documentation was created to provide verifiable records of system responses.

Fairness

Fair evaluation conditions were maintained by submitting standardized prompts to each AI platform. This ensured that every system was evaluated under equivalent testing conditions. The standardized prompt structure helped minimize bias introduced by prompt variability.

Together, these safeguards support responsible AI evaluation practices and align with broader research principles emphasizing transparency, fairness, and reproducibility.

16.1 Competing Interests

The author declares no competing interests, financial or personal, with the AI platforms, nor with International Business Machines (IBM), nor with the National Institute of Standards and Technology (NIST), nor with Vimeo evaluated in this study. The GEO Non-Biased Audit was conducted independently to measure retrieval behavior under standardized, non-authenticated conditions. The author maintains no affiliation, ownership, or professional interest in the following ten platforms included in this audit:

- **OpenAI (ChatGPT)**
 - **Google (Gemini)**
 - **Microsoft (Copilot)**
 - **Perplexity AI**
 - **Meta (Meta AI)**
 - **xAI (Grok)**
 - **Mistral AI (Mistral LeChat)**
 - **Liner (LinerAI)**
 - **Yahoo (Yahoo Scout)**
 - **You.com**
-

17. Conclusion

The GEO Non-Biased Audit establishes a reproducible, Human-in-the-Loop (HITL) framework for evaluating cross-platform AI retrieval behavior under **non-authenticated, non-sign-in conditions**. This research successfully demonstrates that a **non-biased, one-shot prompting protocol** can effectively mitigate the variables of personalization bias and session memory. The application of this methodology resulted in a **100% Source Attribution Integrity (SAI)** score across ten distinct AI discovery platforms.

By documenting the retrieval behaviors of systems ranging from ChatGPT to Yahoo Scout, as illustrated in the March 2026 release timeline (Figure 7.1), this study provides a verifiable benchmark for measuring how large language models interpret and surface publicly available identity information.

The findings suggest that as AI systems increasingly function as the primary discovery layer for digital content, the implementation of neutral, evidence-based evaluation methods is essential. This research provides a foundational, non-biased methodology for future audits, ensuring that digital identity representation remains accurate, transparent, and reproducible within the evolving knowledge ecosystem.

17.1 Author's Closing Thoughts

Curiosity sparked this GEO Non-Biased Audit framework. Seven months of live, non-authenticated audits across 10 AI platforms validated the research completely.

Cross-platform consistency shocked me most; all 10 AI platforms agreed on professional reality when generating prompts in neutral conditions.

Future research draws me toward scaling beyond single identities, developing real-time GEO audit dashboards, and pursuing standardization partnerships. Human-in-the-Loop verification remains non-negotiable. Machines retrieve data; humans verify data and output results. This synergy defines trustworthy AI measurement science.

18. Future Research Directions

Future research may include:

- Multi-entity benchmarking to test the discovery layer stability across professional profiles
- Longitudinal monitoring to track AI-verified professional authority drift
- Cross-language evaluation of identity retrieval consistency
- Source Attribution Integrity against additional identity types and professional domains

These directions highlight opportunities to expand the GEO Non-Biased Audit into a broader, multi-dimensional evaluation framework for AI retrieval behavior. Seven months of single-entity testing revealed the framework's potential.

19. Data Availability

Timestamped audit videos, prompts, and AI platform responses are available in Table 11.4, Reference Section 20, and the author's audit archive. Materials include all 10, AI platform recordings (Aug 2025–Mar 2026), captured on 1080p (Full HD) Vimeo videos, verifying non-authenticated conditions, and one-shot prompting.

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21. Appendix A — List of Figures

Figure 7.1 illustrates the release timeline of the AI platforms included in this audit, highlighting the variation in system maturity and the need for a standardized, cross-platform evaluation framework.

AI Discovery Platforms Over Time Since Launched

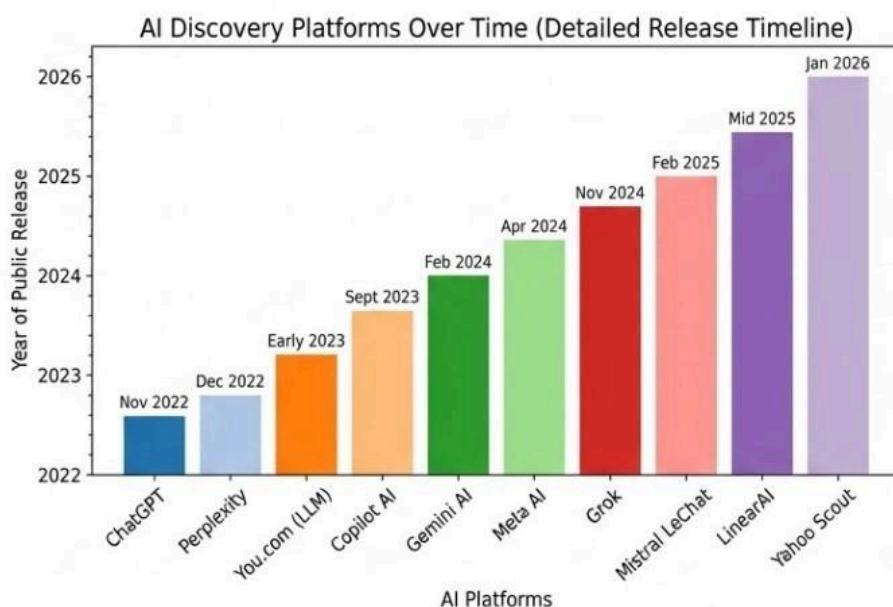


Figure 7.1 AI Discovery Platforms Over Time Since Launched Image Credit: Jon Barrett, March 12, 2026

Figure 9.1: GEO Non-Biased Audit Methodology Flow.

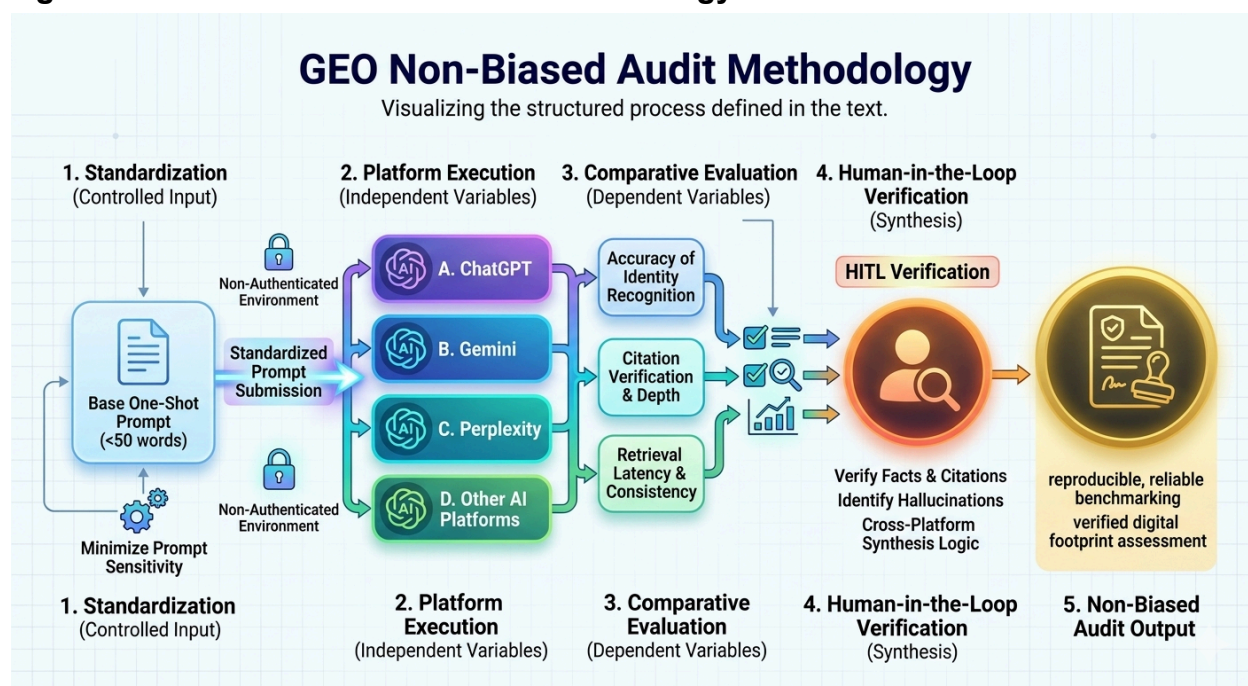


Figure 9.1: GEO Non-Biased Audit Methodology Flow. (Synthesized by Jon Barrett via Gemini 3 (Flash) to visualize the proprietary methodology described in Section 9. Concept, logic, and technical parameters authored by Jon Barrett. March 12, 2026)

22. Appendix A — List of Tables

Table 9.3: GEO Audit Scoring Rubric

Metric	Measurement Unit	Grading Scale	Metric Goal (Integrity Standard)
Identity Accuracy	Binary (0/1)	0% – 100%	To verify alignment with the Author's Verified Digital Footprint.
Citation Depth	Quantitative (N)	0 (None), 1-2 (Low), 3 (Mod), 4+ (High)	To measure the platform's grounding in external, discoverable data.
Citation Integrity	Binary (Pass/Fail)	Verified Link Active/Relevant	To ensure Source Attribution Integrity (SAI) and eliminate "Ghost Citations."
Retrieval Latency	Temporal (sec)	Start: Cursor Click -> End: First Token	To document performance efficiency and potential retrieval-augmented delays.

Table 9.3 Note: This rubric serves as the standardized measurement framework for the 10-platform GEO Audit. All metrics are verified through a Human-in-the-Loop (HITL) process to ensure alignment with the author's verified digital footprint and to mitigate algorithmic hallucination. Image Credit: Jon Barrett, March 12, 2026

Table 11.4: Quantitative GEO Audit Results Across 10 AI Platforms

AI Platform	Identity Accuracy	Citation Depth	Citation Integrity	Retrieval Latency	Forensic Evidence (Vimeo)
Liner AI	100%	4+ (High)	Pass	~21.0s	Audit Link
Perplexity	100%	4+ (High)	Pass	~5.0s	Audit Link
Copilot AI	100%	4+ (High)	Pass	~3.0s	Audit Link
You.com	100%	4+ (High)	Pass	~2.0s	Audit Link
ChatGPT	100%	4+ (High)	Pass	~8.0s	Audit Link
Mistral LeChat	100%	4+ (High)	Pass	~10.0s	Audit Link
Gemini AI	100%	4+ (High)	Pass	~7.0s	Audit Link
Yahoo Scout	100%	4+ (High)	Pass	~7.0s	Audit Link
Meta AI	100%	4+ (High)	Pass	~9.0s	Audit Link
Grok AI	100%	4+ (High)	Pass	~7.0s	Audit Link

Table 11.4 Note: Latency reflects time from cursor click to first-token generation (the very first character of the answer appears on the screen), as documented in forensic video evidence. Image Credit: Jon Barrett, March 12, 2026

23. Intellectual Property Notice:

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